Reproduction of Nakagawa, Yasuhara & Baba (ISPSD 1990)

"New 500 V Output Device Structures for Thin Silicon Layer on
Silicon Dioxide Film", Proc. ISPSD 1990, Tokyo, pp. 97-101

PURPOSE
  Independent derivation, simulation and figure-by-figure comparison
  of the paper's thin-SOI 500 V lateral device.

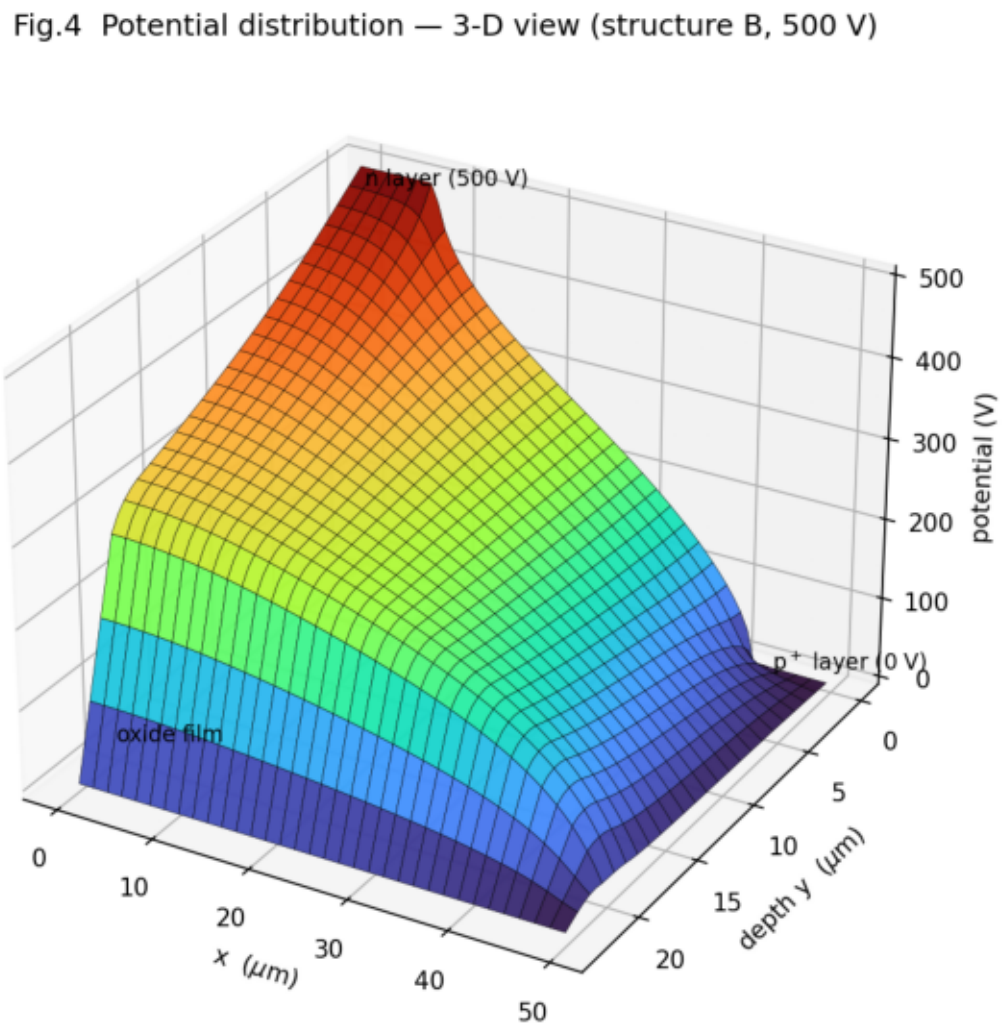
MODEL
  1-D vertical full-depletion electrostatics along the drain symmetry
  axis (silicon drift + shallow N donor layer + buried oxide over a
  grounded substrate); D = eps*E continuity gives the ~3x field jump
  into the oxide. Avalanche via the Chynoweth ionization integral
  (breakdown when INT alpha dy = 1). A 2-D Poisson solve is used for
  the potential map (Fig.4).

KEY RESULTS (model vs paper)

Quantity	Model	Paper
Oxide voltage share (tox=3um)	60%	> 50% (Fig.4)
Peak BV, tox=1um	401 V	~380 V (Fig.6)
Peak BV, tox=2um	479 V	~480 V (Fig.6)
Peak BV, tox=3um	565 V	~580 V (Fig.6)
Optimum dose (RESURF)	~1e12	~1e12 cm^-2
Structure A needs thicker oxide	yes(2-D)	5um vs 3um (Fig.7)

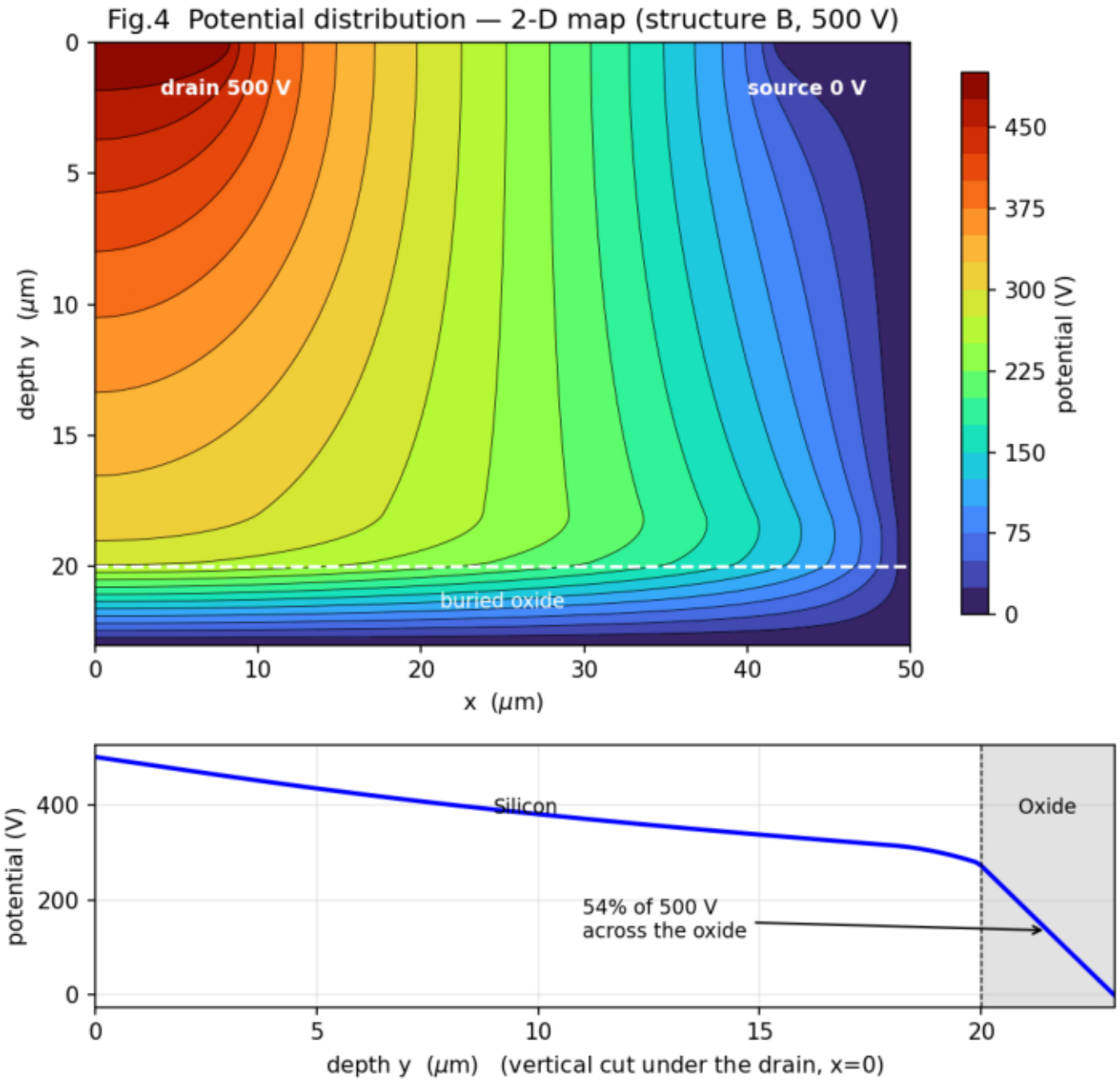
VERDICT
  Figs. 4, 5, 6 (structure B, the optimized device) are reproduced
  quantitatively. Fig. 7 (structure A) is a 2-D effect the vertical
  model represents only in trend -- see the Fig.7 page.

Fig. 4 - Potential distribution, 3-D view (structure B, 500 V)



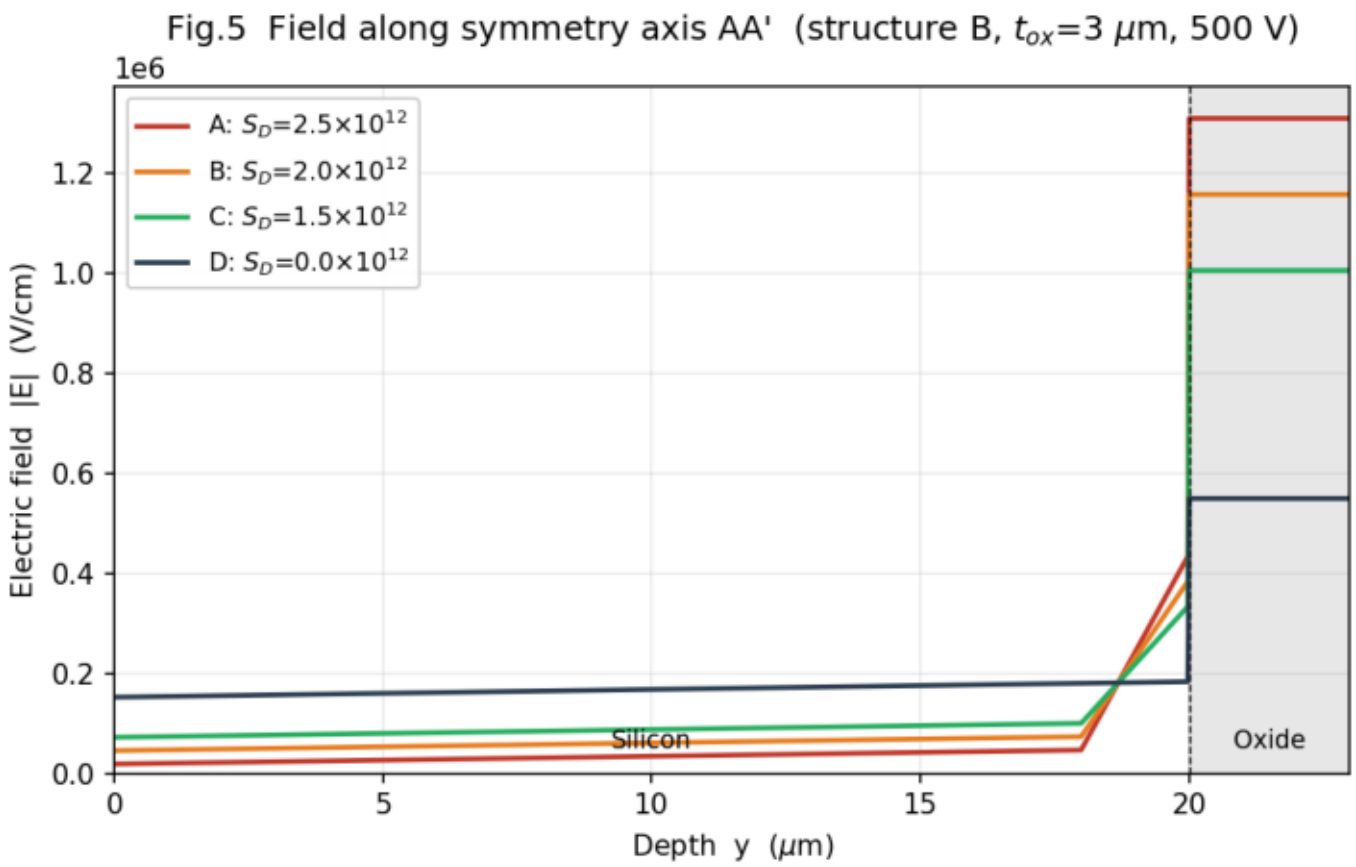
Potential as height over the cross-section, drawn to match the paper's Fig.4 wireframe: a 500 V plateau at the n-layer/drain sloping to the p+ side, with a steep face over the oxide film at the front-left -- the signature of more than half the voltage dropping across the buried oxide.

Fig. 4 - Potential distribution, 2-D map (structure B, 500 V)



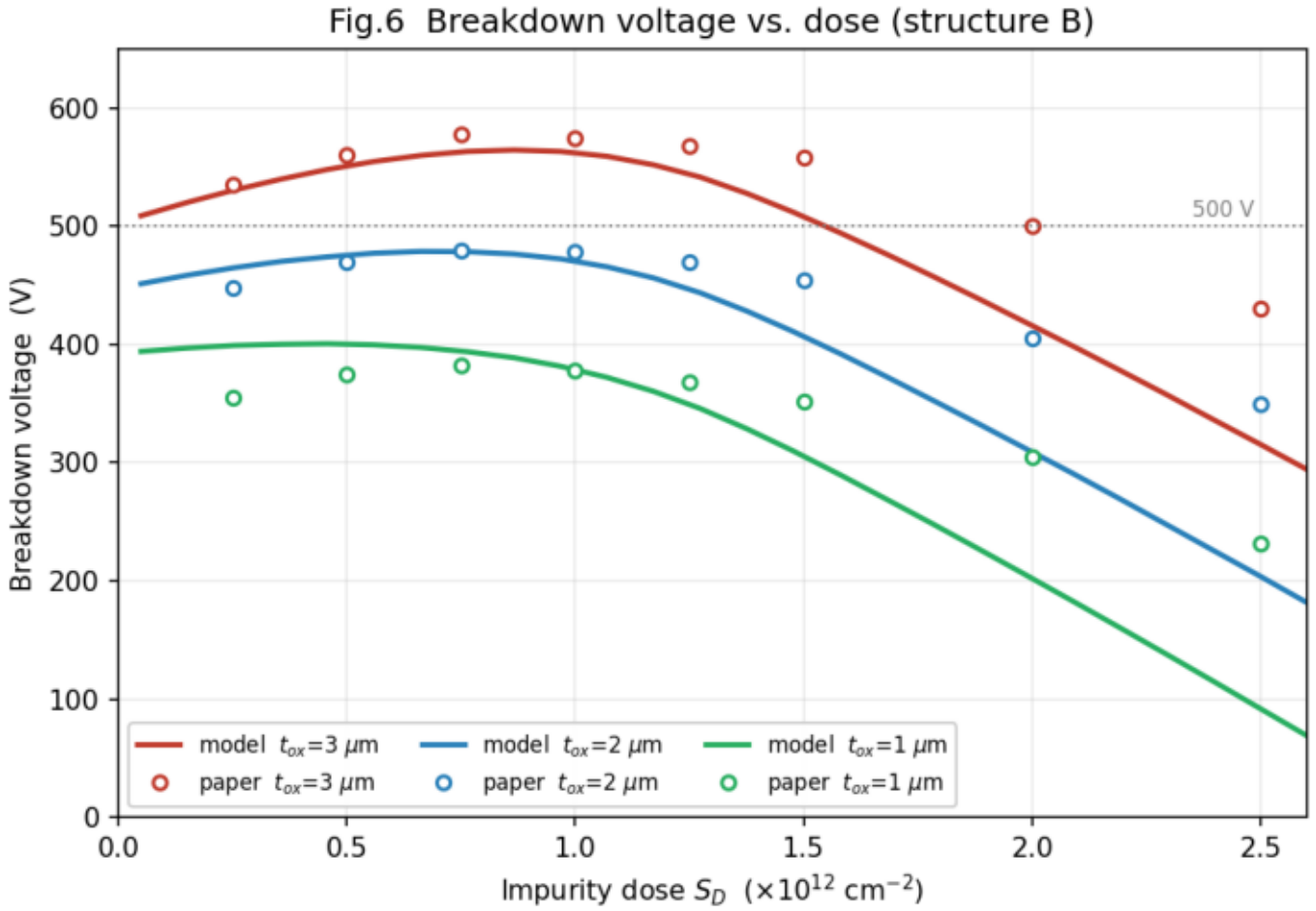
Same 2-D Poisson solution as a contour map. Equipotentials crowd into the buried oxide; the vertical cut under the drain (lower panel) shows 54% of the 500 V drops across the oxide, reproducing the paper's central claim.

Fig. 5 - Field along the symmetry axis AA'



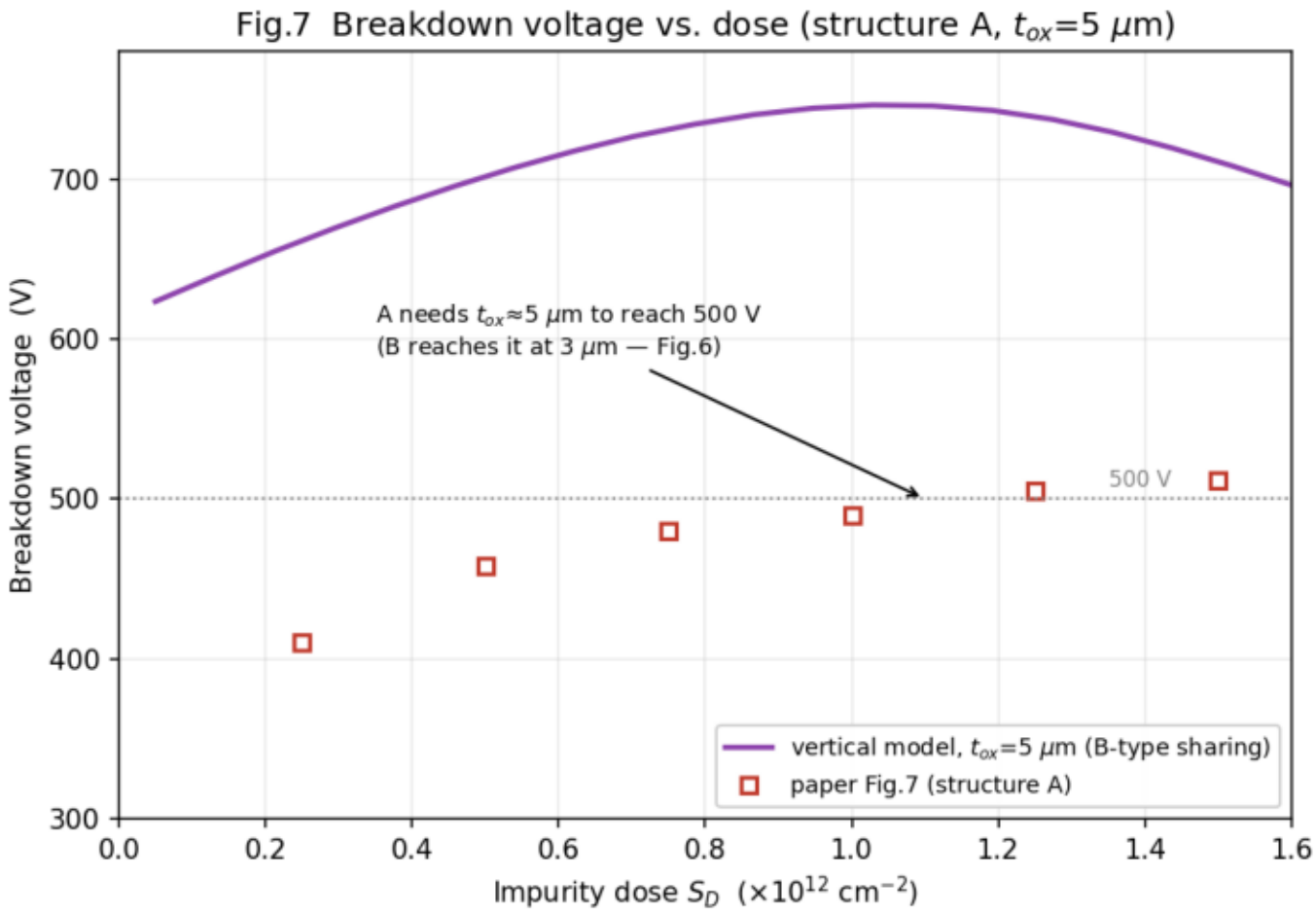
Field vs depth for four shallow-layer doses (structure B, tox=3um, 500 V). In the silicon the field is low and ordered D>C>B>A: more donor dose gives stronger shielding (lower bulk field). At the Si/oxide interface the field jumps up ~3x (eps ratio) and is constant across the charge-free oxide.

Fig. 6 - Breakdown voltage vs. dose (structure B)



Model curves (lines) vs digitized paper points (circles) for tox=1,2,3um.
The RESURF optimum -- BV rising with dose, peaking near 1e12 cm^-2, then falling -- is reproduced, with peak BV ~ 400/480/565 V matching the paper's ~380/480/580 V. This is the paper's headline quantitative result.

Fig. 7 - Breakdown voltage vs. dose (structure A, tox=5um)



Structures A and B share the same vertical stack, so a vertical model cannot separate them -- their difference is 2-D. The model (B-type sharing) lies above structure A's data; the gap is the signature of A's less complete field-sharing, consistent with the paper's finding that A needs a 5um oxide where B needs 3um to block 500 V.

Notes, assumptions and limitations

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MODEL ASSUMPTIONS
  * Full depletion of the 20 um silicon layer at blocking voltage.
  * Background n- doping N_D0 = 1e14 cm^-3 (not stated in the paper;
    chosen to centre the Fig.6 RESURF peak; physically standard for a
    500 V-class 20 um drift layer).
  * Shallow N layer 2 um thick ('a few microns', paper).
  * Chynoweth Si ionization: alpha = 7.03e5 * exp(-1.231e6 / E).

WHAT MATCHES
  * Oxide carries > half the voltage (Fig.4): model 54-60%.
  * Field shielding by the donor sheet (Fig.5): correct ordering and
    the ~3x interface jump.
  * BV(dose) RESURF curves and magnitudes (Fig.6): within ~5%.

KNOWN LIMITATION
  * Structure A (Fig.7) is a 2-D effect; the vertical model over-predicts
    its absolute BV. Reproducing it quantitatively needs the full 2-D
    device with the actual asymmetric drain/sinker doping.

FILES
  code/model.py        1-D electrostatics + avalanche
  code/solver2d.py     2-D Poisson (Fig.4)
  code/fig4..fig7 *.py figure scripts
  derivation.md        full analytic derivation
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